HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 05 Name: _____ Marks: / 102

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Simplify $\frac{(uv^2)^5}{u^7v^{-3}}$ and express your answer with positive indices. (3 marks) (DSE-2026 PI Q02)

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2. Factorize

(a) $5x^2y + 17xy - 12y$,

(b) $5x^2y + 17xy - 12y - 10x + 6$.

(4 marks) (DSE-2026 PI Q04)

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3. The sum of the number of blue pens and the number of red pens in a box is 795. If 123 blue pens are taken out from the box, then the number of blue pens in the box is 25% less than the number of red pens in the box. Find the number of red pens in the box. (4 marks) (DSE-2026 PI Q05)

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4. In Figure 1, ABC is a triangle. D is a point lying on AB. Let E and F be points lying on CB produced and CD produced respectively such that $BD//EF$ and $BC = EF$. It is given that $BD = 8$ cm and $BE = 6$ cm.

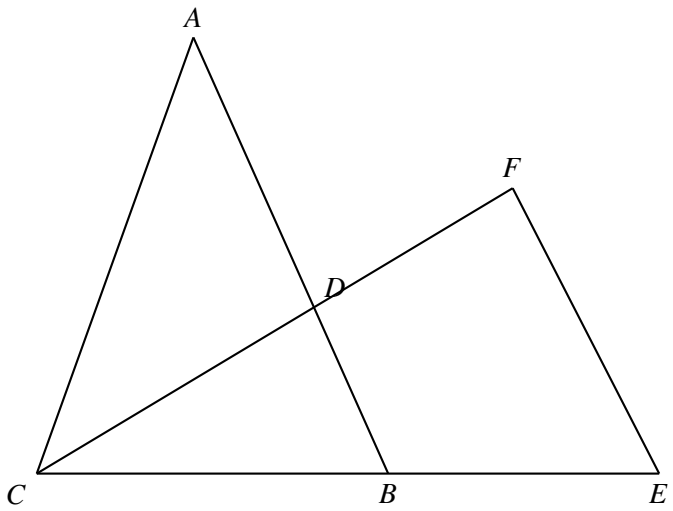


Figure 1

(a) By considering a pair of similar triangles, find the length of BC.

(b) Suppose that $AD = 10$ cm. Prove that $\triangle ABC \cong \triangle CEF$.

(5 marks) (DSE-2026 PI Q08)

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5. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

Stem (tens)	Leaf (units)
2	5 6 9
3	2 3 3 6 6 7 8
4	0 0 2 2 5 5 7 7 7
5	w

The ratio of the range to the inter-quartile range of the distribution is 7 : 3.

(a) Find w.

(b) (i) Write down the mode of the distribution.

(ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution.

(6 marks) (DSE-2026 PI Q10)

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6. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

(a) Find the equation of L.

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K.

(i) Find the coordinates of P.

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(7 marks) (DSE-2026 PI Q12)

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7. A solid metal sphere of radius 10 cm is melted and recast into three similar solid right circular cylinders X, Y and Z. Let r_1 cm, r_2 cm and r_3 cm be the base radii of X, Y and Z respectively. It is given that $r_1 : r_2 : r_3 = 2 : 3 : 5$.

(a) Express the volume of Y in terms of π.

(b) A craftsman finds that the height of Y is 9 cm.

(i) Find r_2.

(ii) The craftsman claims that the total surface area of Z is greater than the surface area of the sphere. Do you agree? Explain your answer.

(8 marks) (DSE-2026 PI Q13)

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8. When the polynomial $g(x)$ is divided by x^2+2x+a, the quotient and the remainder are $x+13$ and $b(x-3)$ respectively, where a and b are constants. It is given that $g(x)$ is divisible by $x-3$.

(a) Find a.

(b) Is there only one possible value of b such that at least two roots of the equation $g(x)=0$ are equal? Explain your answer.

(8 marks) (DSE-2026 PI Q14)

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9. A bag contains 3 red balls, 3 white balls and 5 black balls. If 4 balls are randomly drawn from the bag at the same time, find

(a) the probability that exactly 2 red balls are drawn;

(b) the probability that exactly 2 red balls or exactly 2 white balls are drawn.

(4 marks) (DSE-2026 PI Q15)

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10. It is given that P, Q and R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of PR is $3x+5y-27=0$. Let C be the circle which passes through P, Q and R.

(a) Find the equation of C.

(b) The tangent to C at P and the tangent to C at R intersect at the point H. Is $\triangle HPR$ a right-angled triangle? Explain your answer.

(6 marks) (DSE-2026 PI Q16)

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11. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face VAB of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

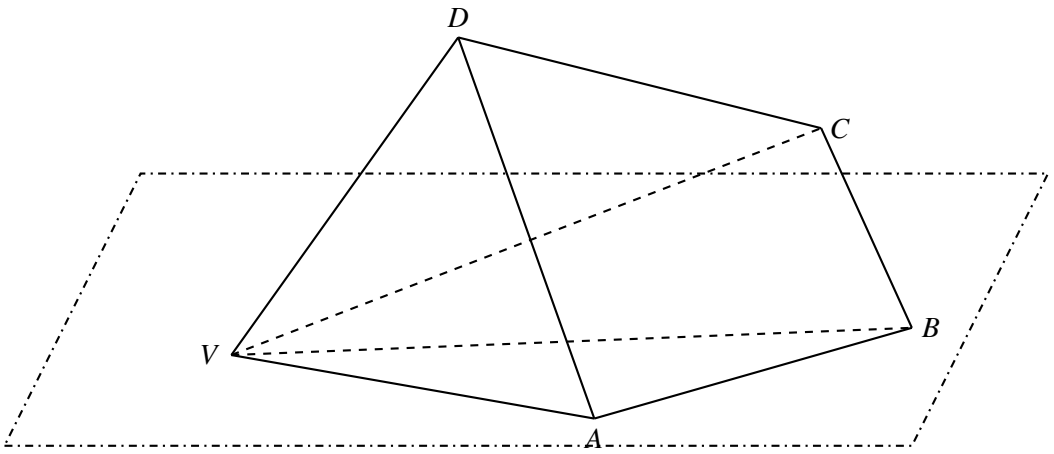


Figure 2

(a) Find

(i) the distance between A and C,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between VD and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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12. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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13. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4-x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

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14. If m and n are constants such that $(x-5)(x+m)-4(x+n)=x(x+6)+1$, then $n=$

A. -19.

B. -3.

C. 9.

D. 15.

(DSE-2026 PII Q05)

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15. Solve $(x+2s)(x+t) = sx+st$, where s and t are constants.

A. $x = -s$

B. $x = -2s$

C. $x = -s$ or $x = -t$

D. $x = -2s$ or $x = -t$

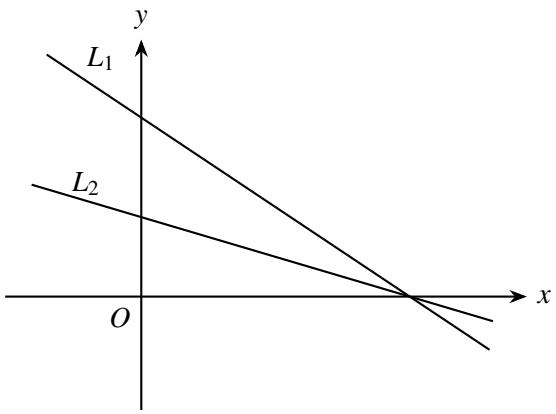
(DSE-2026 PII Q06)

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16. In the figure, the equations of the straight lines L_1 and L_2 are $ax+y=b$ and $cx+y=d$ respectively. It is given that L_1 and L_2 intersect at a point on the positive x-axis. Which of the following are true?



I. $a < c$

II. $b > d$

III. $ad = bc$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2026 PII Q14)

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17. The perimeter and the area of a sector are k cm and $3k$ cm^2 respectively. If the radius of the sector is 14 cm, then the arc length of the sector is

A. 19 cm.

B. 21 cm.

C. 28 cm.

D. 49 cm.

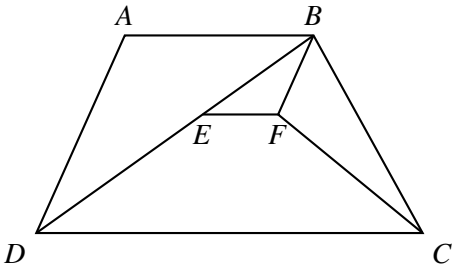
(DSE-2026 PII Q16)

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18. In the figure, $ABCD$ is a trapezium with $AB//DC$ and $AB : CD = 3 : 5$. Let E be a point lying on BD such that $BE : ED = 2 : 3$. It is given that $AB//EF$ and $AD//BF$. If the area of $\triangle ABD$ is 225 cm^2, then the area of the trapezium $CDEF$ is



A. 222 cm^2.

B. 260 cm^2.

C. 279 cm^2.

D. 285 cm^2.

(DSE-2026 PII Q18)

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19. It is given that $\triangle PQR$ is a right-angled triangle with $\angle QPR = 90^\circ$. Denote the area of $\triangle PQR$ by S. If $QR = p$, $PR = q$ and $\angle PQR = \theta$, which of the following must be true?

A. $S = \frac{1}{2}pq$

B. $S = \frac{1}{2}pq\sin\theta$

C. $S = \frac{1}{2}pq\cos\theta$

D. $S = \frac{1}{2}pq\tan\theta$

(DSE-2026 PII Q19)

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20. It is given that $WXYZ$ is a quadrilateral, where $\angle WXY = \angle YWZ$ and $\angle WYX = \angle WZY$. If $WX = 9$ cm, $WY = 40$ cm and $XY = 41$ cm, find the perimeter of the quadrilateral $WXYZ$.

A. 100 cm

B. 180 cm

C. 410 cm

D. 490 cm

(DSE-2026 PII Q21)

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21. It is given that $ABCDEF$ is a regular 6-sided polygon. If AC and BF intersect at the point G, which of the following are true?

I. $BG//CE$

II. $\triangle ABG \sim \triangle BDC$

III. $\triangle AGF \cong \triangle BGC$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2026 PII Q23)

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22. Which of the following can be obtained from any box-and-whisker diagram?

I. Mean

II. Lower quartile

III. Inter-quartile range

A. I only

B. II only

C. I and III only

D. II and III only

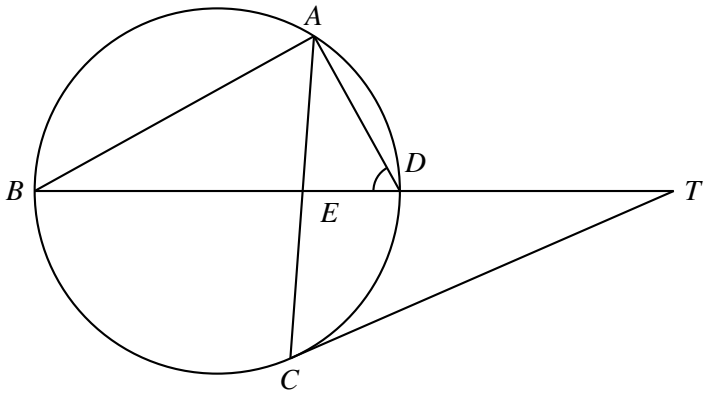
(DSE-2026 PII Q29)

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23. In the figure, $ABCD$ is a circle. AC and BD intersect at the point E. BD is produced to the point T such that TC is tangent to the circle. It is given that $AB//TC$. If $\angle AED = 85°$ and $\angle CTE = 23°$, then $\angle ADE =$



A. 39°.

B. 56°.

C. 62°.

D. 72°.

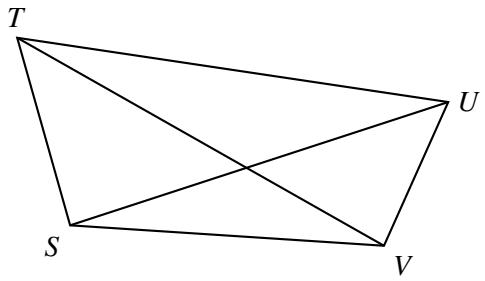
(DSE-2026 PII Q38)

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24. The figure shows the quadrilateral $STUV$, where $ST = 8$ cm and $SV = 5$ cm. If $\angle UTV = 30^\circ$ and $\angle TSU = \angle TVU = 90^\circ$, find $\angle SUV$ correct to the nearest degree.



A. 22°

B. 28°

C. 30°

D. 38°

(DSE-2026 PII Q39)

I can finish _____ questions

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